

DRP Summer 26 Writing Sample

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The following question is from my Math 465 Homework 2.

Problem. Determine the number of possible choices of sets A, B such that

$$\emptyset \subseteq A \subseteq B \subseteq \{1, 2, \dots, n\}.$$

Solution. To first start this problem we have to understand that the size of A depends on the size of B . Given that we want to pick k -elements out of $[n]$ to form B and we want to go through all options of $k \Rightarrow k \in [0, n] \Rightarrow \binom{n}{k}$ ways to pick B , but now that B has a fixed size $k \Rightarrow A$ has 2^k possibilities. Then by the multiplication rule

$$\Rightarrow \sum_{k=0}^n \binom{n}{k} 2^k \text{ total combinations.}$$

But we can simplify by the Binomial Theorem

$$\Rightarrow \sum_{k=0}^n \binom{n}{k} 2^k = (2 + 1)^n = 3^n$$

total options of subsets.

Reflection. This was my original solution to the problem and as I arrived at that result I was kind of surprised that it was so clean and concise. But as I thought about it more it makes total sense which led me to this second method of proving the problem. Given an element of $[n]$ there exists 3 possible states for that element to be in. 1 - The element is not in A nor B , 2 - The element is in B , but not A , or lastly 3 - The element is in both A and B , thus a sort of “Trinary String” emerges where each position has 3 states $\Rightarrow 3^n$ possibilities. This solution is fascinating to me since it parallels exactly to my professor’s original proof of how a singular subset of $[n]$ has 2^n possibilities using binary strings where each position has 2 states either is a member of the subset or is not!

This problem highlights one of my favorite things about math: that one problem can have many different pathways that all arrive at the same solution. But it also secretly highlights that some solutions carry more intuition and explanations than others. Take my first solution which wasn’t very illuminating as to why the answer is 3^n , but simply arrives there using a lot of heavy lifting from the binomial theorem which obscures a great deal of intuition. But that second approach highlights how the solution is directly connected to how we originally calculated the number of subsets of $[n]$!